HDT · 12 standard cases — User manual (HTML tool)

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About the tool

HDT · 12 standard cases checks flexible pavement designs to IRC:37-2018 for Cases 101–112, on a phone or computer, with nothing to install. It works out stresses and strains itself (layered elastic theory), so no other program is needed. This manual covers version v4.14.

- **For:** quick design checks and comparisons by engineers who already know IRC:37-2018.
- **Checks:** εt (fatigue of the bituminous layer), εv (rutting of the subgrade), and for some cases εt CTB, σt under the dumper and εv on the GSB.
- **Accuracy:** results agree with IITPAVE within about 0.1–0.3%; this tool reads slightly higher.
- **Companion:** the HDT Excel workbook (HDT_V008.xlsm) runs the same rows through IITPAVE for confirmation.

Disclaimer. This tool is an aid for quick checks only. It does not replace IITPAVE, the analysis software of IRC:37-2018. Final designs must be checked with IITPAVE and approved by a qualified engineer. The user is responsible for all inputs and results.

Getting started

Open the link in **Chrome** (Android) or **Safari** (iPhone), get a key once from the tool owner, and the tool unlocks on that phone from then on.

1. Open the link you received: https://manishplusanand.github.io/flex-pav-quick-irc37.html. If the page stays on "Loading…", open it in Chrome or Safari instead of a file viewer.
2. The lock screen shows your **Device ID** (like HDT-AB12-CD34-EF56-GH78). Tap **Send on WhatsApp** or **Copy**, and send it to the tool owner.
3. You receive a key starting with **HDT1.** Copy the whole key.
4. Paste it into the box and tap **Unlock**.
5. The tool opens. The bottom of the page shows *Licensed to <your name> · this phone only · valid until <date>*.

The key is saved on the phone, so next time the tool opens directly.

- **One key, one phone.** A key works only on the phone whose Device ID it was made for.
- **Expiry.** If your key has an end date, ask the owner for a new key when it expires.
- **New versions** open with your existing key; just open the same link.

The four tabs

The tool has four tabs across the top; the red-edged disclaimer box sits above them on every screen.

Tab	What it is for
Dashboard	Counts of rows, passes and fails; a table with the worst row of each case; the list of all design rows, which can show all rows, defaults only or user-added only. Tap any row to open it.
Design	Pick a row, change its inputs and see its results; print, copy, delete or make it a default.
Records	Save your rows to a file, open a file, export to Excel, see every row, manage the default list, and bring back deleted rows.
Notes	How the numbers are worked out, the points read, the 12 cases, references and the disclaimer.

The **About the 12 cases** section (Dashboard and Notes) and **About Case …** (Design) open and close with a tap.

Working with design rows

Each design is a **row**; the tool starts with 12 default rows, one per case, and you can add as many of your own as you like.

Each row's name is built from its values, for example *★ 1. Case 101 · 100 msa · CBR 8% · 40 BC + 140 DBM + 250 WMM + 250 GSB*. It updates as you edit.

To check a design

1. Open **Design** and pick a row in the top box.
2. Change the white input boxes on the left (thicknesses, CBR, traffic, moduli and so on). Results update about half a second after you stop typing.
3. The **Worked out from inputs** list under the inputs shows derived values such as subgrade MR and the permissible strains.

Buttons in Design

- **Copy as new row** — makes a copy just below; use it to try a variation without losing the original.
- **Delete row** — tap twice to confirm. At least one row always stays. A deleted row can be brought back from Records → Recently deleted.
- **Make default / Update default** — adds the row to the default list, or saves its new values there.
- **Copy summary** — copies a short text summary for a message or email.
- Changing the **Case** box switches the row to another case; inputs that case needs are filled with standard values.

The default list (★)

- Rows in the default list show ★ at the start of their name.
- The list holds **at most 12 rows**. When it is full, remove one in **Records** before adding another.
- **Records → Reset my rows to the default list** brings your rows back to the defaults (tap twice). **Restore the 12 standard defaults (list and rows)** (tap twice) puts the original 12 back into the list and adds any of them missing from your rows. Your other rows are kept.
- An edited ★ row keeps its ★; the saved default keeps the old values until you tap **Update default**.

All rows, defaults only or user-added only

- **Records → All rows** lists every design row; ★ marks the defaults.
- Three buttons filter the list: **All rows**, **Defaults only** and **User-added only**, each with its count. The same buttons are on the Dashboard under **All design rows**. The tool remembers your choice.
- Each row has **Open**, and either **Make default** or **Remove from defaults**. Removing from defaults does not delete the row.
- "User-added" means any row that is not in the default list.

Bringing back a deleted row

1. Go to **Records → Recently deleted**. It keeps your last 50 deleted rows.
2. Tap **Restore** on a row, or **Restore all**. Restored rows go to the end of your rows.

Empty this list (tap twice) clears it for good.

Reading the results

A row **passes** when every check's actual value is at or below its permissible value; the largest **actual ÷ permissible** ratio is the **governing** ratio.

Verdict cards (Design). One card per check, showing the actual value, the permissible value, the ratio, and the life back-calculated in msa. The bar's black mark is at 100%.

Colour bands (same as the Excel workbook):

Actual ÷ permissible	Colour	Meaning
above 100%	dark red	Exceeds permissible – FAILURE
95% to 100%	yellow	Very close to limit
90% to 95%	orange	Within limit
85% to 90%	green	Satisfactory margin
below 85%	pink	Substantially below limit

Analysis tables. Under **Analysis runs** the tool prints the stresses and strains at each point, laid out like an IITPAVE output (Z, R, SigmaZ … epR). Rows marked **L** are just below an interface, in the lower layer. As in IRC:37-2018, positive values are tensile and negative values compressive; only the absolute value (without + or −) is used in the performance models (Eq. 3.1 to 3.6).

- **Boxed cells** are the values compared for a check, as in IRC:37-2018 Fig. I.3.
- The **shaded, bold** cell is the value used: the largest of the boxed cells by size, ignoring the sign.
- For εt, the tool compares epT and epR in the "L" row under the wheel (R = 0) and the upper row between the wheels (R = 155), and takes the largest of the four by size, ignoring the sign.
- For εv, it compares epZ in the two "L" rows at the top of the subgrade and takes the larger by size.

Borderline rows. The tool reads about 0.15% higher than IITPAVE, so a row right at 100% may show FAIL here and PASS in IITPAVE. Confirm such rows with the HDT Excel workbook.

Printing reports

You can print one row or all rows, on paper or as a PDF, in A4.

1. In **Design**, tap **Print this row** or **Print all rows**. **Print all rows** is also on the Dashboard.

2. A print preview opens. Untick **Include analysis tables** to leave out the stress and strain tables and save pages.
3. Tap **Print / Save PDF**. To get a PDF, choose **Save as PDF** as the printer.
4. Tap **← Back** to return to the tool.

- **One row:** a 3-page report with design checks, pavement section, inputs, method, disclaimer and signature boxes; the analysis tables follow on a new page.
- **All rows:** a summary page with every row's ratios and PASS/FAIL, then one report per row, each starting on a new page. For 12 rows, about 37 pages with analysis tables or 25 without.
- **Licence line:** every report shows *Licensed to <name> · this phone only …* at the top and in the footer. Reports and Excel exports show the name only; the key and Device ID are never printed.
- For Cases 103 and 104 the report carries the CTB fatigue damage note; for Cases 105–112 it names IRC:SP:59-2019.

If no print screen opens on a phone, use the browser menu → Share → Print.

Saving and moving your records

Your rows are saved automatically in the browser on that phone, but a saved file is the only safe copy, so save one regularly.

Save a copy (Records → Save your records)

1. Tap **Save records to file**. A file named like *HDT records 2026-11-08 1430.json* goes to your Downloads.
2. Keep it somewhere safe, or send it to yourself.

Open a saved file

1. Tap **Open records file** and choose the .json file.
2. Choose **Add to my rows** to keep your current rows and add the file's rows, or **Replace my rows** (tap twice) to switch to the file's rows and default list.

If downloads don't work on your phone, tap **Copy records as text** and paste the text into a note or email.

Rows can disappear if the browser's data is cleared, or if you open the tool in a different browser or on a different phone. A saved file brings them back.

Export to Excel for an IITPAVE check

Records → Export rows to Excel (Design Input format) makes an Excel file of all your rows, laid out exactly like the Design Input sheet of the HDT Excel workbook, so the same designs can be run in IITPAVE.

1. Tap **Export rows to Excel**. A file named like *HDT rows 2026-11-08 1430.xlsx* goes to your Downloads.
2. On a computer, open it next to HDT_V008.xlsm.
3. In the export, select whole rows 7 onward (click the row numbers) and copy.
4. In HDT_V008.xlsm, go to **Design Input**, click row number 7 and paste.
5. Run from the workbook's Dashboard.

The export has two sheets:

- **Design Input** — your rows with the workbook's own formulas; the result columns are left empty for IITPAVE to fill.
- **v4 results** — this tool's strains, deflection, ratios and PASS/FAIL for each row, with matching row numbers, so you can compare with IITPAVE.

The workbook has room down to row 200 (about 190 rows).

The 12 cases in brief

Cases 101–104 use IRC:37-2018 moduli; Cases 105–112 add geosynthetic reinforcement by the MIF or LCR method of IRC:SP:59-2019.

Case	Layers	Method	Checks
101	BC + DBM + WMM + GSB	Standard	εt, εv
102	BC + DBM + WMM + CTSB	Standard	εt, εv
103	BC + DBM + AIL + CTB + CTSB	Standard	εt, εv, εt CTB, σt dumper
104	BC + DBM + AIL + CTB + GSB	Standard	εt, εv, εt CTB
105	BC + DBM + WMM(R) + GSB	MIF	εt, εv
106	BC + DBM + WMM(R) + GSB(R)	MIF	εt, εv, εv GSB
107	BC + DBM + WMM(R) + CTSB	MIF	εt, εv
108	BC + DBM + WMM + GSB(R)	MIF	εt, εv, εv GSB
109	BC + DBM + WMM(R) + GSB	LCR	εt, εv

Case	Layers	Method	Checks
110	BC + DBM + WMM(R) + GSB(R)	LCR	εt, εv
111	BC + DBM + WMM(R) + CTSB	LCR	εt, εv
112	BC + DBM + WMM + GSB(R)	LCR	εt, εv

BC bituminous concrete · DBM dense bituminous macadam · AIL aggregate interlayer · WMM wet mix macadam · CTB cement-treated base · GSB granular subbase · CTSB cement-treated subbase · (R) reinforced with geosynthetic. The **About the 12 cases** section in the tool explains each case in two or three lines.

CTB fatigue damage (Cases 103, 104). IRC:37-2018 Cl. 3.6.3 and 11.1.7 (Eq. 3.6 and 3.7) also require a cumulative fatigue damage (CFD) analysis of the CTB, using the axle load spectrum from an axle load survey, 0.80 MPa contact pressure and the CTB's 28-day flexural strength. **This tool does not do it.** Carry it out separately; the CTB is acceptable only if CFD is less than 1.0.

Geosynthetics (Cases 105–112). MIF and LCR values must come from field and laboratory investigations or validated manufacturer data, as IRC:SP:59-2019 requires.

Reliability. All checks use the 90% reliability equations, as IRC:37-2018 Cl. 3.7 requires for Expressways, National Highways, State Highways and Urban Roads.

Troubleshooting and common questions

Problem	What to do
Page stays on "Loading…"	Open the link in Chrome (Android) or Safari (iPhone), not in a file viewer or an app's built-in browser.
"This key is for another phone"	Each key works on one phone. Send this phone's Device ID to the owner for its own key.
"This key expired"	Ask the owner for a new key.
"Not a valid key"	Copy the whole key exactly, from HDT1. to the end, with nothing added.
The tool asks for a key again	Your browser's data was cleared. Paste your key again. If the Device ID has changed, send the new one to the owner.

Problem	What to do
My rows are gone	If you deleted them, restore them from Records → Recently deleted. Otherwise, open your last saved file with **Records → Open records file**.
A row shows an error	An input is blank or zero. Check every white box for that case.
Dashboard count is not what I expect	Each line of the Dashboard's case table shows the worst row of that case; the full list is under **All design rows**.
Nothing happens on Print	Use the browser menu → Share → Print.
Result differs slightly from IITPAVE	Expected: about 0.1–0.3% higher here. Rows near 100% should be confirmed in IITPAVE.

References

- IRC:37-2018, *Guidelines for the Design of Flexible Pavements* (Fourth Revision), Indian Roads Congress, November 2018.
- IRC:SP:59-2019, *Guidelines for Use of Geosynthetics in Road Pavements and Associated Works* (First Revision), Indian Roads Congress, August 2019.